Indian Institute of Technology Gandhinagar

Course No.: CS 328 **Course Title:** Introduction to Data Science

Credits (L T P C): 3 1 0 4

Prerequisites (if any): None

Expected frequency of offering: Every year

Google Classroom Link: https://classroom.google.com/c/ODQ4NzU4NzUyNTUx?cjc=vxvys6dh

Course contents

Foundations of Data Representation: Overview of data representation (as vectors, sets, graphs); basics of feature creation and **high-dimensional embeddings**; distance measures; probability basics, central limit theorem, random variables, and concentration inequalities.

Dimensionality Reduction & Spectral Methods: Singular Value Decomposition (SVD)/PCA and best-fit subspaces; low-rank approximation; eigenvectors and power method; **Vector Databases and similarity search (LSH/FAISS)**; tSNE and other low-dimensional embeddings.

Probabilistic Modeling & Estimation: Modeling and estimation; maximum likelihood (MLE), Bayesian estimators; mixture models (gaussians, document-topic models); **Expectation-Maximization (EM) intuition and data imputation.**

Inference & Hypothesis Testing: Sampling methods (e.g. unbiasing, bootstrapping, confidence intervals); hypothesis testing basics and augmentations (e.g. non-parametric, multiple hypothesis testing, Bayes Factor); introduction to causality, counterfactuals, and **A/B testing (bucket testing).**

Large Scale Graph Analytics: Clustering (k-means/median, k-means++, Lloyd's algorithm); graph analytics: ranking methods (pagerank, HITS) and connection to random walks; community detection (expansion, conductance, modularity) and connection to eigenvalues; triangle counting.

Big Data Paradigms & Streaming: Introduction to memory efficient approximate computation, Bloom filters, bit arrays; Streaming model: samples and sketches (reservoir sampling, Morris' algorithm, Count-Min sketch, AMS sampling); **Logic of Distributed paradigms (MapReduce).**

Ethics, Privacy & Fairness: Ethical and privacy issues in data science; introduction to privacy models (e.g. Differential Privacy); **quantifying algorithmic fairness and bias in data-driven systems.**

Learning outcomes

At the end of the course, students will be able to:

- Understand the statistical and algorithmic ideas that underlie the field of data science.
- Extract patterns from medium to large data sets using algorithms for clustering and dimensionality reduction.
- Formulate and argue about statistical significance using hypothesis testing and bootstrapping.
- Design memory-efficient sketches and distributed logic (MapReduce) for big data streams.
- Utilize practical tools (NumPy, SciPy, Matplotlib, Scikit-learn) for data analytics and visualization.

Grading policy

- **Problem Sets / Assignments / Tutorials:** 15% — The problem sets will be given to solve. Delayed submission will have penalties.
- **Project (with mandatory Report):** 20%
- **Tests (2 total):** 60% — Mid semester and End semester examination.
- Attendance: Attendance will have a weightage of 5%.

Make-up tests will not be offered except under exceptional circumstances with valid documentation, and late submissions of assignments may be subject to penalties.

Texts/References

Foundational Books:

- *Foundations of Data Science* by Avrim Blum, John Hopcroft, and Ravi Kannan.
- *Python for Data Science Handbook* by Jake Vanderplas. (Available Online)
- *Computational and Inferential Thinking* by Ani Adhikari and John DeNero. (Available Online)
- *Advanced Data Analysis from an Elementary Point of View* by Cosma Shalizi. (Available Online)

Supplementary Papers & Open Resources (Modern Topics):

- *Fairness and Machine Learning* by Barocas, Hardt, and Narayanan. (Full PDF at fairmlbook.org)
- *MapReduce: Simplified Data Processing on Large Clusters* by Jeffrey Dean and Sanjay Ghemawat.
- *Equality of Opportunity in Supervised Learning* by Moritz Hardt et al. (NIPS '16).
- *Billion-scale similarity search with GPUs* by Jeff Johnson et al. (For Vector Search logic).

Tentative weekly plan

Week	Topics
Week 1	Foundations: Data representation (vectors, sets, graphs), feature creation, and Data-Centric AI concepts.
Week 2	Probability Recap: Random variables, Central Limit Theorem (CLT), and concentration inequalities.
Week 3	Hashing: Hashing basics, balls and bins, MinHash and Locality Sensitive Hashing (LSH) applications.
Week 4	Dimensionality Reduction I: Singular Value Decomposition (SVD), PCA, and best-fit subspaces.
Week 5	Dimensionality Reduction II: Low-rank approximations, power method, and Vector Database indexing (FAISS logic).
Week 6	Clustering: k-means, k-center, Lloyd's algorithm, and k-means++.
Week 7	Graph Analytics I: PageRank, HITS, and connection to random walks.
Week 8	MID-SEM BREAK

Week 9	MID-SEM BREAK
Week 10	Graph Analytics II: Community detection, expansion, conductance, and spectral algorithms.
Week 11	Latent Variable Models: Gaussian Mixture Models (GMM) and Expectation-Maximization (EM) algorithm intuition.
Week 12	Big Data Summaries: Memory-efficient computation, Bloom filters, and bit arrays.
Week 13	Streaming & Sketches: Reservoir sampling, Count-Min sketch, and Morris' algorithm for approximate counting.
Week 14	Distributed Paradigms: The MapReduce logic, data locality, and handling data at scale.
Week 15	Statistical Inference: Sampling, estimation, confidence intervals, and bootstrapping.
Week 16	Hypothesis Testing & Experiments: Multiple hypothesis testing, Bayes Factor, and A/B Testing (Bucket Testing).
Week 17	Ethics & Fairness: Algorithmic bias, fairness metrics (Equality of Opportunity), and Differential Privacy.

Any other remarks

Nil.